Research Archives and the Cost of Price Discovery

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Abstract

We study how institutions' choices from a research archive affect information aggregation through order flow. In a strategic trading model, institutions select mixtures of forecasts generated by useful and nuisance learning and may purchase a service that reduces nuisance exposure. The dealer observes the chosen signal laws and prices aggregate order flow. With correlated errors, a symmetric interior archive equilibrium lies beyond the price-information maximum and before the standalone-information maximum. Unequal learning rates create a two-dimensional archive, whose relevant deployment choices lie on its boundary. Correction changes both signals and archive choice. Institutions may purchase correction at costs for which a planner purchases none, even when both sides reoptimize deployment. This expenditure difference reflects the gap between trading rents and the real-investment value of price information. It can persist with independent errors, when private archive choices are informationally efficient in the two-institution benchmark. The model links the information selected for trading to price discovery and the resources devoted to improving it.

Keywords: price discovery; informed trading; information acquisition; research archives; correlated errors.
JEL classification: G14, D82, D83.

1 Introduction

Price discovery depends on which private signals informed institutions choose to trade on. A research archive makes that choice endogenous: it contains forecasts whose useful information and exposure to common errors have developed at different rates. An institution can select a version, combine versions, or purchase correction of its chosen forecast. These decisions change both its trading profits and the information conveyed by aggregate order flow. We ask how privately chosen deployment differs from deployment that maximizes the investment value of market prices.

Costly information earns a private return through trading opportunities (Grossman & Stiglitz, 1980). Information acquisition and market prices must therefore be determined jointly (Verrecchia, 1982). A retained archive adds a composition choice to acquisition: the institution selects among information structures generated by the same research technology. The relevant tradeoff concerns how much useful information accompanies each unit of residual and common error. Standalone forecast accuracy captures only part of this tradeoff.

We place this archive choice in a strategic trading market with competing informed institutions and a competitive dealer. A bounded learning rule generates versions with useful and nuisance loadings that develop at different rates. Before future innovations arrive, institutions publicly choose a nonnegative mixture of versions and whether to purchase a correction service. Each receives one private forecast. Institutions anticipate how their deployment changes order-flow informativeness and price impact. A real investor uses the market price to choose investment. The archive's production cost is sunk; correction consumes additional real resources.

Archive choice separates the private value of a forecast from its contribution to prices. Along a one-dimensional learning path, the marginal improvement in residual precision competes with rising nuisance exposure. With positive error correlation and the stated path conditions, a symmetric interior pure equilibrium lies beyond the price-information maximum and before the standalone-information maximum. The institution partly values differentiation from its rival, yet tolerates more common error than is desirable for price discovery. Unequal learning rates extend this choice to a two-dimensional convex set. For two institutions, an outward radial improvement places private best responses and efficient deployments on the boundary, so at most two versions implement each relevant archive choice.

Paid correction then changes the versions institutions find attractive. In the numerical illustration, the private uncorrected archive combines versions four and five, while correction shifts deployment to version seven. The planner chooses different mixtures. Both institutions can prefer correction at a cost for which the planner purchases none, even after each side reoptimizes its archive. This expenditure effect reflects the familiar difference between private and social returns to information. The archive determines the information structures over which those returns are evaluated. The two effects are distinct: with independent nuisance errors, private deployment is informationally efficient in the two-institution model, while correction expenditure can still be excessive.

The market foundation is Park (2026), which studies correlated processing errors, technology adoption and quadratic real-investment welfare. We study endogenous private-information choice within that market: institutions choose which part of a produced research archive to use for trading. The learning path yields the private archive-ordering result, and its geometry permits a joint comparison of deployment and paid correction. Strategic prediction-rule competition can generate collectively suboptimal outcomes in the arti-

ficial stock market of Joshi et al. (2002). Here institutions choose from a common research technology and anticipate how each deployment changes strategic trading and the price.

Research technology also shapes information acquisition in Dugast & Foucault (2018) and Dugast & Foucault (2025). In the former, cheaper early low-precision information can crowd out more precise research and reduce eventual price informativeness. In the latter, data abundance raises price informativeness even when search effort falls. Our model focuses on useful and nuisance learning within a completed archive. Their relative speeds determine the attainable information compositions, and trading incentives determine which composition is deployed.

Common information has different individual and collective values. Economies of information production can induce common purchases and comovement (Veldkamp, 2006a, 2006b). Coordination motives can make the welfare effect of public information ambiguous (Morris & Shin, 2002). An algorithm that performs better in isolation can also lower collective decision quality (Kleinberg & Raghavan, 2021). In our setting, common nuisance exposure makes improvements in one institution's forecast contribute less to information aggregation. The archive embeds that exposure in the same choice as useful learning.

Retained research experience motivates the deployment margin. FactorMiner uses accumulated research memory to guide financial-factor exploration (Wang et al., 2026). The model captures a stylized tradeoff in such retained research products through its two coefficient paths. The learning mechanism acts before trading: the institutions and dealer subsequently optimize with a common understanding of the signal law. This timing differs from reinforcement-learning speculators that sustain supra-competitive trading profits (Dou et al., 2025) and market makers that learn their pricing strategies (Colliard et al., 2026).

The analysis proceeds from research technology to archive choice, then correction expenditure. The final section examines how access to several future version outputs changes the information available from the same archive. Complete proofs accompany the article. The computational supplement reports the numerical methods, additional calculations and environment tables.

2 Research and the trading market

An archive is a set of information technologies available for deployment. Its versions have already been produced, but the future signal realizations have not arrived. An institution chooses how to combine those versions and whether to purchase a correction service. These commitments determine the information structure that the dealer prices. This timing follows the distinction between buying an information structure and buying an already realized signal emphasized by Bergemann & Bonatti (2019).

2.1 A research technology with two learning rates

Repeated evaluation can reward both useful learning and sample-specific fit. Suppose an evaluation sample contains orthogonal feature vectors u and z, with $u'u = z'z = n$ and $u'z = 0$. Expanding the squared error of the prediction $xu + wz$ gives the improvement criterion

$$\hat{q}x + \hat{b}w - \frac{x^2 + w^2}{2}, \qquad \hat{q} = \frac{u'y}{n}, \quad \hat{b} = \frac{z'y}{n}. \tag{1}$$

A nuisance feature can have a nonzero sample projection even when its population projection is zero. Optimizing a noisy model-selection criterion can consequently overfit that criterion (Cawley & Talbot, 2010).

Adaptive reuse of evaluation data creates a related distinction between evaluation performance and generalization (Dwork et al., 2015). We study the technology conditional on $\hat{q} = 1$ and $\hat{b} = b \geq 0$.

Quadratic adjustment costs generate a bounded learning path. Subtract $(x - x_{t-1})^2/(2h_x) + (w - w_{t-1})^2/(2h_w)$ from equation (1), where both learning rates are positive. The maximization is a strictly concave quadratic problem, which gives

$$x_t = \frac{x_{t-1} + h_x}{1 + h_x}, \qquad w_t^0 = \frac{w_{t-1}^0 + h_w b}{1 + h_w}, \qquad x_0 = \tfrac{1}{2}, \quad w_0^0 = 0. \tag{2}$$

This is a proximal update: the new criterion is optimized with a penalty for moving from the preceding coefficients (Parikh & Boyd, 2014). The two rates are coordinates of the same update. They allow useful learning to mature at a different speed from the accumulation of nuisance exposure. The explicit solution is

$$x_t = 1 - \tfrac{1}{2} r_x^t, \qquad w_t^0 = b(1 - r_w^t), \qquad r_x = (1 + h_x)^{-1}, \quad r_w = (1 + h_w)^{-1}. \tag{3}$$

The history traces an attainable tradeoff between useful and nuisance loadings. Useful learning raises the fundamental component relative to residual noise; nuisance learning raises exposure to an irrelevant future shock. Adaptive-analysis bias provides a statistical motivation for the second component (Russo & Zou, 2016). The financial signal law specifies how both components enter the trading market.

Institution i chooses weights α_i over the produced versions $t = 1, \ldots, T$. The coefficients at $t = 0$ initialize research. The institution also chooses $v_i \in \{0, 1\}$, where $v_i = 1$ purchases correction at real cost $c \geq 0$. The delivered coefficients are

$$a_i = \sum_{t=1}^{T} \alpha_{it} x_t, \qquad w_i = \frac{\sum_{t=1}^{T} \alpha_{it} w_t^0}{1 + v_i}, \qquad \alpha_{it} \geq 0, \quad \sum_{t=1}^{T} \alpha_{it} = 1. \tag{4}$$

The correction service halves nuisance exposure while preserving useful loading and residual variance. The cost c is both the institution's expenditure and the real resources used by the service. Write $\mathcal{H}_v$ for the convex hull of $(x_t, w_t^0/(1 + v))$. Every point of this hull is available under equation (4); the archive's production cost is sunk.

2.2 Public commitment to one output

The deployment contract delivers one private scalar forecast after future innovations arrive. Institutions publicly commit to nonnegative archive weights and correction status. A licensed service that delivers a specified forecast provides an interpretation of this contract. The chosen weights specify a signal law with residual variance d. Access to several future version realizations is a separate information arrangement, examined in Section 6.

Public deployment choices let the dealer price the information structure while signal realizations remain private. Strategic sharing of realized information can itself change price impact and trading incentives (Goldstein et al., 2025). Here the public object is the chosen signal law, which the institutions and dealer understand.

The risky asset's terminal value is $v \sim N(0, V)$, where $V > 0$. For $N \geq 2$ institutions, the delivered signals are

$$Y_i = a_i v + w_i \zeta_i + \sqrt{d}\,\epsilon_i, \qquad \mathrm{Var}(\zeta_i) = \tfrac{1}{2}, \quad \mathrm{Cov}(\zeta_i, \zeta_j) = \tfrac{\kappa}{2} \quad (i \neq j). \tag{5}$$

Here $d > 0$ and $0 \leq \kappa \leq 1$. The residuals ϵ_i are independent standard normal variables. They are independent of v and the jointly Gaussian nuisance shocks, which are also independent of v. The covariance matrix is

$$\Sigma = Vaa' + \frac{\kappa}{2} ww' + \mathrm{diag}\left(d + \frac{1-\kappa}{2} w_i^2\right). \tag{6}$$

Positive residual variance makes Σ positive definite. The nuisance correlation κ concerns errors conditional on the fundamental. It is different from the correlation of the signals, which also share the fundamental. The market inherits the common-error formulation and real-investment interpretation of Park (2026). Archive choice endogenizes the coefficients that enter that formulation.

2.3 Informed trading and price formation

Institutions value information through the trades it supports. This link between information choice and investment choice also appears in portfolio information acquisition (Van Nieuwerburgh & Veldkamp, 2010). Here institutions are risk-neutral strategic traders. Institution i submits $X_i = \beta_i Y_i$, and independent noise orders $u_0 \sim N(0, U)$ have variance $U > 0$. A competitive risk-neutral dealer observes $Q = \sum_i X_i + u_0$ and sets $p = \mathbb{E}[v \mid Q]$. We work within the linear Gaussian trading equilibrium.

The trading stage is a multi-informed version of the strategic market in Kyle (1985). Correlated private signals determine both competition and information revelation in the models of Foster & Viswanathan (1996) and Back et al. (2000). The matrix solution below accommodates unequal deployed archives. Let $D_\Sigma = \mathrm{diag}(\Sigma)$, $g = Va$ and $H = \Sigma + D_\Sigma$.

Proposition 1 (Trading equilibrium)**.** *For the maintained signal law and $U > 0$, the linear trading equilibrium with positive price impact satisfies*

$$k = H^{-1} g, \qquad \lambda = \sqrt{\frac{k' D_\Sigma k}{U}}, \qquad \beta = \frac{k}{\lambda}, \qquad p = \lambda Q. \tag{7}$$

Institution i earns gross expected profit

$$\Pi_i = \frac{\Sigma_{ii} k_i^2}{\lambda}, \qquad J := \mathrm{Var}(\mathbb{E}[v \mid p]) = g' k. \tag{8}$$

Choosing a different archive changes both the information in an institution's order and the price impact it faces. The covariance matrix, trading intensities and dealer's pricing rule adjust together. This equilibrium response determines the private return to deployment. Appendix A derives the trading solution.

Real investment gives price information an economic value. The distinction between informational and economic efficiency is central to Dow & Gorton (1997), while Bond et al. (2012) emphasizes information that is useful for real decisions. Add a price-taking investor who chooses an incremental investment $I \in \mathbb{R}$ after observing p. Its payoff is $vI - \gamma I^2/2$, where $\gamma > 0$. The optimal choice is $I(p) = \mathbb{E}[v \mid p]/\gamma$, with expected value $J/(2\gamma)$.

Trading rents are transfers in this welfare account. The counterparties' losses offset the informed institutions' gross trading profits, and the competitive dealer earns zero expected profit. The fundamental is exogenous to the real investment decision. Net incremental social value is consequently

$$W = \frac{J}{2\gamma} - c\sum_{i=1}^{N} v_i. \tag{9}$$

The planner values correction through the investment improvement and subtracts its resource cost. This same account governs the archive and expenditure comparisons below.

3 Archive choice and price discovery

A useful forecast and an informative market price value different error reductions. The distinction can be expressed with two normalized error components, after which the archive geometry determines which combinations are attainable. This connects the information comparison to the institution's actual choice of research versions.

3.1 Information along a research path

Normalize a symmetric signal by its positive useful loading. Define residual error $E = d/a^2$ and nuisance error $P_e = w^2/(2a^2)$. The normalized variance and covariance are

$$S = V + E + P_e, \qquad K = V + \kappa P_e. \tag{10}$$

The units are fundamental-variance units. Increasing a reduces residual error relative to useful information. Increasing w/a raises nuisance error in those same units. The price-information formulas therefore depend on composition as well as total error.

An individual forecast, the market price and a pool of forecasts contain different information. Gaussian projection and Proposition 1 give

$$J_{\text{solo}} = \frac{V^2}{V + E + P_e}, \tag{11}$$

$$J_{\text{price}} = \frac{NV^2}{(N+1)V + 2E + [2 + (N-1)\kappa]P_e}, \tag{12}$$

$$J_{\text{pool}} = \frac{NV^2}{NV + E + [1 + (N-1)\kappa]P_e}. \tag{13}$$

Each quantity is a variance of a conditional mean, so a larger value means that more fundamental uncertainty is resolved. The pooled quantity is a hypothetical statistic available to an observer of every signal. Prices instead aggregate strategic orders through noisy order flow. The difference has the same conceptual origin as the distinction between dispersed information and its noisy price aggregate in Diamond & Verrecchia (1981). Here imperfect competition contributes an additional wedge between pooling and prices.

Suppose a differentiable archive path is indexed by z. Assume $E'(z) < 0$, $P'_e(z) > 0$, and that $\theta(z) = P'_e(z)/[-E'(z)]$ is strictly increasing. The ratio θ is the marginal nuisance increase per unit of residual-error reduction. Differentiating the denominators in equations (11)–(13) gives the information-maximizing cutoffs

$$\theta_{\text{pool}} = \frac{1}{1 + (N-1)\kappa}, \qquad \theta_{\text{price}} = \frac{2}{2 + (N-1)\kappa}, \qquad \theta_{\text{solo}} = 1. \tag{14}$$

When the path crosses these cutoffs and $\kappa > 0$, the pooled optimum precedes the price optimum, which precedes the standalone optimum. Common nuisance exposure carries a greater cost for aggregation than for an isolated forecast. The standalone signal can keep improving after the price has passed its information maximum. This comparison is about the use of a given information technology. It does not yet specify what a profit-maximizing institution chooses.

Information composition can matter beyond the present scalar fundamental. In Goldstein & Yang (2015), information diversity creates complementarities through different payoff components. Coordination motives also affect the acquisition of common information (Hellwig & Veldkamp, 2009). Our path holds the payoff dimension fixed and tracks how a retained history changes residual and nuisance errors together. That restriction makes the marginal tradeoff in equation (14) explicit.

3.2 A private equilibrium cutoff

A private archive deviation changes one institution's signal law and reprices the market. At a symmetric profile, write

$$A = 2S - K, \qquad D = 2S + (N-1)K. \tag{15}$$

Both quantities are positive on the maintained domain. When only institution i moves along the path, its own normalized variance changes by $E' + P'_e$. Each covariance involving that institution changes by $\kappa P'_e/2$. The factor one half appears because only one of the two nuisance loadings changes.

Differentiating the matrix solution yields

$$\frac{d \log \Pi_i}{dz_i} = -G_M(E' + P'_e) - G_K \frac{\kappa P'_e}{2}, \tag{16}$$

where

$$G_M = \frac{4(D-K)}{AD} - \frac{2}{ND} - \frac{2N-1}{2NS}, \tag{17}$$

$$G_K = \frac{2(N-1)(D-2K)}{AD} - \frac{2(N-1)}{ND}. \tag{18}$$

The coefficients measure the profit costs of own variance and cross-institution covariance. The following identities establish their signs:

$$G_K = \frac{2(N-1)[(N-1)A + N(N-2)K]}{NAD} > 0, \tag{19}$$

$$(N-1)G_M - G_K = \frac{(N-1)[2S + (2N-1)K]}{2NSA} > 0. \tag{20}$$

Thus $G_M > 0$, and positive correlation gives

$$\theta_{\text{price}} < \theta_{\text{private}} := \frac{G_M}{G_M + \kappa G_K/2} < 1. \tag{21}$$

Proposition 2 (Location of a symmetric interior archive equilibrium)**.** *Let $N \geq 2$ and $\kappa > 0$. Suppose a feasible scalar archive path has $E' < 0$, $P'_e > 0$ and strictly increasing θ. Suppose the price and standalone cutoffs are crossed on the path. Every symmetric interior pure equilibrium on that path satisfies*

$$z_{\text{price}} < z_{\text{NE}} < z_{\text{solo}}. \tag{22}$$

The institution partly values differentiation from its rival and therefore stops before the standalone-information maximum. Its trading return nevertheless tolerates more nuisance accumulation than the price-information objective. Proposition 2 locates a symmetric interior equilibrium under the stated path conditions. At $\kappa = 0$, the strict cutoff separation disappears; archive endpoints can also bind.

Equal learning rates give an exact scalar example. If $h_x = h_w = h$, define $z_t = 1 - (1 + h)^{-t}$. Every archive has coefficients

$$a = \tfrac{1}{2}(1 + z), \qquad w = bz, \qquad z = \sum_t \alpha_t z_t \in [z_1, z_T] \tag{23}$$

before correction. Set $b = 1$, $h = 1/20$, $T = 8$, $\kappa = 4/5$, $d = 1/10$ and $V = U = 1$. Then $\theta(z) = 5z$. The price maximum is $z = 1/7$, the private profile is $z \simeq 0.179277482$, and the standalone maximum is $z = 1/5$. Price information equals 0.540540541 at its maximum and 0.539858625 at the private profile.

3.3 From the scalar path to the full archive

Unequal learning rates create a second archive dimension. For the first three uncorrected versions,

$$\det \begin{pmatrix} 1 & x_1 & w_1^0 \\ 1 & x_2 & w_2^0 \\ 1 & x_3 & w_3^0 \end{pmatrix} = \tfrac{1}{2} b\, r_x r_w (r_x - 1)(r_w - 1)(r_w - r_x). \tag{24}$$

Positive b and unequal positive learning rates make the determinant nonzero. A single effective archive age is then insufficient to describe all attainable coefficients. Any hull point can be implemented with at most three versions by affine dependence. A boundary point can be implemented with at most two versions. These are representations of the feasible set, which leave every original mixture available.

A dominance argument reduces the complete two-dimensional problem to its boundary. For two institutions and $V = U = 1$, let s and t be their normalized variances, and let c_0 be their normalized covariance. Define

$$D_0 = 4st - c_0^2, \quad L = 2t - c_0, \quad M = 2s - c_0, \quad Q_0 = sL^2 + tM^2. \tag{25}$$

The first institution's profit is $sL^2/(D_0\sqrt{Q_0})$. Holding the rival variance and covariance fixed gives

$$\frac{\partial \log \Pi_i}{\partial s} = -\frac{2c_0^2 Q_0 + sD_0(L^2 + 4tM)}{2sD_0Q_0} < 0, \tag{26}$$

$$\frac{\partial J}{\partial s} = -\frac{2(2t - c_0)^2}{D_0^2} < 0, \tag{27}$$

provided $L, M > 0$. These inequalities hold throughout the archive domain used for computation. Indeed, $a_i \geq 1/2$ and $0 \leq w_i \leq 1$ imply $0 \leq \rho_i := w_i/a_i \leq 2$. For $0 \leq \kappa \leq 1$,

$$2S_i - K_{ij} = 1 + 2E_i + \rho_i^2 - \frac{\kappa}{2}\rho_i\rho_j \geq 1 - \frac{\kappa^2\rho_j^2}{16} \geq \frac{3}{4}. \tag{28}$$

Proposition 3 (Boundary reduction)**.** *Consider two institutions with $V = 1$, $d > 0$, $a_i \geq 1/2$, $0 \leq w_i \leq 1$ and $0 \leq \kappa \leq 1$. A strictly interior point of a two-dimensional feasible archive is neither a private best response nor part of a price-information-maximizing archive allocation.*

An outward radial move raises (a_i, w_i) proportionally. It preserves nuisance exposure relative to the fundamental, but reduces d/a_i^2. At an interior point, a sufficiently small such move is feasible. Equations (26) and (27) then improve both objectives. The argument applies after correction as well, since the attenuation remains inside the same coefficient bounds. A one-dimensional hull must be solved over its entire segment; its relative interior is not excluded by this proposition.

Boundary deployment retains the choice of intermediate mixtures. When useful learning matures quickly, an institution can combine adjacent versions to gain useful loading at a lower nuisance cost than a later version would impose. The efficient mixture also accounts for how that nuisance exposure enters the rival's forecast. The numerical illustration below shows both choices within the same attainable archive.

Independent errors give a useful benchmark. For $\kappa = 0$, normalized covariance is fixed at $V = 1$, and each institution's archive affects its normalized variance through $e_i = (d + w_i^2/2)/a_i^2$. Equations (26) and (27) imply that both private profit and price information improve as e_i falls. Conditional on correction status, private and efficient archives therefore minimize the same feasible error. Correlation is what separates their deployment objectives in this model.

4 Correction expenditure and real resources

A correction service changes the information on which an institution trades. Its private return is the resulting increase in expected trading profit. The planner instead values the investment improvement net of resources. Efficient use of information does not imply efficient acquisition, as Colombo et al. (2014) shows. The archive makes this distinction operational: both the purchaser and the planner can change which versions they deploy after selecting the correction technology.

4.1 Deployment and correction choices

Fix the two-institution model and write $\Pi(r_i, r_j)$ for gross profit, where $r_i = (a_i, w_i)$ is a deployed archive. Let r_0 be a symmetric archive equilibrium when both institutions have $v_i = 0$, and let r_1 be one when both have $v_i = 1$. The cost c is constant within either regime, so these conditional archive problems maximize gross profit. Define the willingness-to-switch boundaries

$$C_{00} = \sup_{r \in \mathscr{R}_1} \Pi(r, r_0) - \Pi(r_0, r_0), \tag{29}$$

$$C_{11} = \Pi(r_1, r_1) - \sup_{r \in \mathscr{R}_0} \Pi(r, r_1). \tag{30}$$

The first boundary measures the gain from adding correction against an uncorrected rival. The second measures the smallest profit loss from abandoning correction against a corrected rival. Each supremum allows a simultaneous archive change and uses the resulting dealer repricing. Consequently, $(r_0, 0)$ at both institutions is an equilibrium of the joint choice game when $c \geq C_{00}$. The profile $(r_1, 1)$ at both institutions is an equilibrium when $0 \leq c \leq C_{11}$.

The switching thresholds reflect a unilateral decision: the rival retains its deployment while the institution chooses its best archive under the alternative correction status. The conditional archive equilibrium rules out changes that preserve status.

Information-acquisition complementarities can produce nontrivial equilibrium sets (Ganguli & Yang, 2009). Here the thresholds identify the costs supporting each symmetric regime. The result below concerns costs that support both institutions purchasing correction.

4.2 The global planner's cost envelope

The planner can allocate correction asymmetrically and reoptimize both archives. Let

$$J_m^* = \max_{\substack{v_1+v_2=m \\ r_i \in \mathscr{R}_{v_i}}} J(r_1, r_2), \qquad m \in \{0, 1, 2\}. \tag{31}$$

The entire cost problem is the upper envelope

$$W^*(c) = \max_{m \in \{0,1,2\}} \left\{ \frac{J_m^*}{2\gamma} - mc \right\}. \tag{32}$$

For a fixed number of corrected forecasts, cost leaves the efficient archive unchanged. The three information values therefore determine the planner's choice for every nonnegative cost.

Zero correction is strictly efficient above an explicit threshold. Comparing the zero-correction line with each other line in equation (32) gives

$$C_P = \max \left\{ \frac{J_1^* - J_0^*}{2\gamma}, \frac{J_2^* - J_0^*}{4\gamma} \right\}. \tag{33}$$

For $c > C_P$, every efficient allocation uses zero corrections. At the threshold, correction counts can tie. The computed threshold uses upper bounds for J_1^* and J_2^* and an attainable lower bound for J_0^*, so it preserves the strict comparison.

A separate threshold compares the two feasible private archive profiles. Let $J_{00} = J(r_0, r_0)$ and $J_{11} = J(r_1, r_1)$, and define

$$C_W = \frac{J_{11} - J_{00}}{4\gamma}. \tag{34}$$

At $c > C_W$, correction at the private archives lowers net value relative to the feasible uncorrected profile. This comparison gives the value loss directly; the planner threshold separately determines efficient expenditure.

Proposition 4 (Excessive correction expenditure)**.** *Suppose r_0 and r_1 are conditional symmetric archive equilibria, and $C_{11} > \max\{C_W, C_P, 0\}$. At every cost satisfying*

$$\max\{C_W, C_P, 0\} < c < C_{11}, \tag{35}$$

both institutions purchasing correction at r_1 is an equilibrium of the joint archive-and-correction game. The global planner selects zero corrections. The corrected equilibrium also has lower net value than the feasible uncorrected profile at r_0.

Archive reoptimization enters both sides of the expenditure comparison. The institution can change deployment when it switches correction status. The planner can assign different archives to the two institutions and purchase just one correction. The interval in equation (35) supports excessive expenditure after these adjustments.

The underlying resource tension has precedents in trading technology. In Biais et al. (2015), private investment in fast trading exceeds the social optimum because institutions do not internalize its externalities. Budish et al. (2015) studies rents that sustain an arms race for speed. In the present model, the purchased technology reduces an information error and can increase J. Excess expenditure arises when its private trading value supports a cost above its real-investment value.

4.3 Trading rents and the cost region

Noise orders scale the private incentive without changing price information. The vector k in equation (7) is independent of U, whereas λ is proportional to $U^{-1/2}$. Hence

$$\Pi_i(U) = \sqrt{U}\,\Pi_i(1), \qquad J(U) = J(1). \tag{36}$$

Multiplying every gross profit by the same positive constant leaves each conditional archive best response unchanged. The cost boundaries therefore satisfy $C_{00}(U) = \sqrt{U}\,C_{00}(1)$ and $C_{11}(U) = \sqrt{U}\,C_{11}(1)$. The real-value boundaries C_W and C_P do not depend on U. If $C_{11}(1) > 0$, a nonempty interval in Proposition 4 exists whenever

$$U > \left[\frac{\max\{C_W, C_P, 0\}}{C_{11}(1)}\right]^2. \tag{37}$$

Noise trading raises the returns an institution can capture from an improved forecast while leaving the investor's information gain fixed. For the economy in Section 5, $C_W > C_P$ and the threshold is $U \simeq 0.145622$. The relative size of private and social returns also depends on γ: the social thresholds scale with $1/\gamma$, so their comparison with private willingness to pay is governed by $\gamma\sqrt{U}$.

The value of information depends on how equilibrium behavior uses it (Angeletos & Pavan, 2007). In this model, correlation creates the strict deployment-ordering inequality. The expenditure comparison instead reflects trading rents relative to real investment value, and can persist when nuisance errors are independent.

5 Quantitative implications

The numerical economies show how learning rates affect deployed forecasts and the value of correction. Institutions choose within the same archive as the planner. Their different objectives determine which versions enter prices and how much correction they purchase.

5.1 Benchmark economies

The numerical domain varies the strength and timing of nuisance learning. It contains the Cartesian product

$$b \in \{0, \tfrac{1}{2}, 1\}, \qquad h_x, h_w \in \{\tfrac{1}{20}, \tfrac{1}{5}, \tfrac{1}{2}\}, \qquad \kappa \in \{0, \tfrac{1}{5}, \tfrac{4}{5}\}. \tag{38}$$

The benchmark has two institutions and eight deployable versions. We set $V = U = \gamma = 1$ and $d = 1/10$. The initial loadings are $(x_0, w_0) = (1/2, 0)$, and correction halves nuisance exposure. These values specify the economic benchmark, including the relative scale of trading rents and investment value. Equal learning rates produce a line segment; $b = 0$ removes nuisance learning. In that case, h_w and κ do not affect the outcome, and the 27 parameter combinations represent three distinct learning paths.

In each economy, we compute private archive choices with neither or both institutions purchasing correction. The planner chooses archives for correction profiles $(0, 0)$, $(0, 1)$ and $(1, 1)$, allowing asymmetry in every case. The profile $(1, 0)$ has the same planner value as $(0, 1)$ by relabeling institutions.

Numerical equilibria have a unilateral-gain bound of 10^{-10} over the full feasible archive, including switches in correction status at the stated costs. Planner values have an error bound of 10^{-8}. The computational supplement gives the methods and the full table of parameter combinations.

5.2 Which versions institutions deploy

An unequal-speed economy illustrates the economic role of archive choice. Set $b = 1$, $h_x = 1/2$, $h_w = 1/20$ and $\kappa = 4/5$. Useful learning advances quickly, while nuisance learning accumulates gradually. Figure 1 plots all eight versions and their convex hull. Every nonnegative mixture is available. The private uncorrected archive lies between versions four and five; its weight on version five is approximately 0.665582. The efficient uncorrected allocation uses version four at both institutions. Moving farther along the lower boundary improves the useful loading, but it also raises nuisance exposure that the private trader values differently from the real investor.

Correction changes which history the institution finds attractive. The private corrected profile deploys version seven. The efficient allocation with two corrections lies between versions six and seven. Thus the comparison allows an institution to exploit the corrected technology by changing its archive. Table 1 reports the deployed coefficients for the private profiles and the planner's allocations. The table includes the allocation with one corrected institution. The useful and nuisance coordinates both enter the trading equilibrium; a single average archive age would miss these differences.

The uncorrected private profile yields $J_{00} = 0.605577266$. The efficient archive raises this value by 0.000406233, approximately 0.067 percent of the efficient information level. Private correction raises price information to $J_{11} = 0.616650170$. The archive loss and the resource comparison thus concern different margins: the first is the composition of deployed information, while the second includes the cost of improving it.

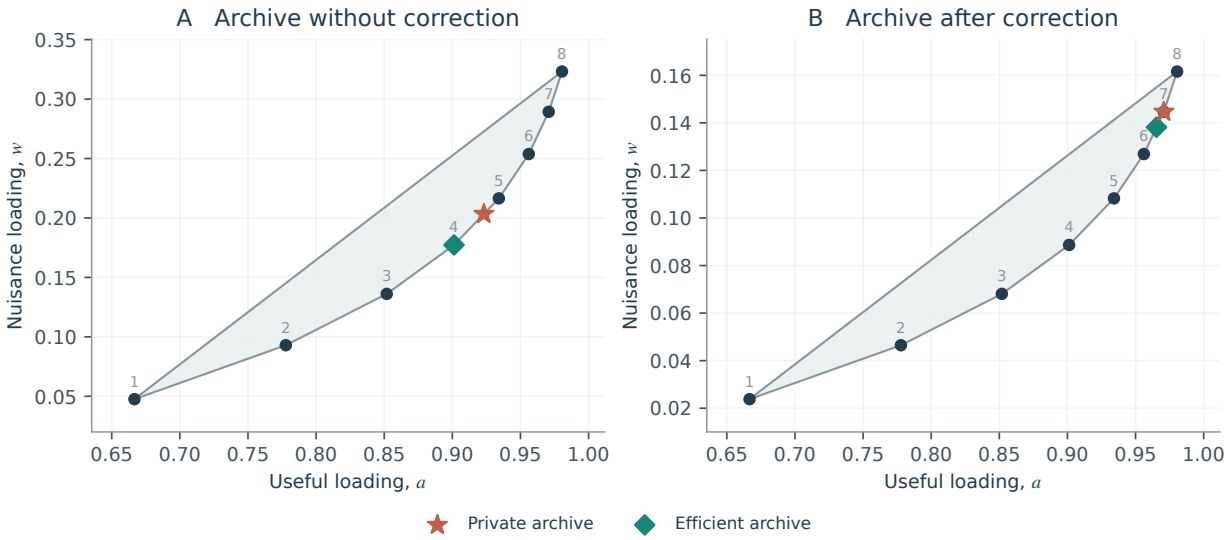


Figure 1. Archive deployment before and after correction. The displayed economy has $b = 1$, $h_x = 1/2$, $h_w = 1/20$ and $\kappa = 4/5$. Dots label the eight research versions; the shaded polygon is their complete convex hull. Stars denote private symmetric archive profiles, and diamonds denote efficient archives conditional on equal correction status. Efficient profiles happen to be symmetric in this example.

Table 1. Deployed archives in the unequal-speed illustration.

Allocation	a_1	w_1	a_2	w_2
Private, no correction	0.923147	0.203373	0.923147	0.203373
Planner, $(0, 0)$	0.901235	0.177298	0.901235	0.177298
Planner, $(0, 1)$	0.901235	0.177298	0.956104	0.126892
Planner, $(1, 1)$	0.965375	0.138150	0.965375	0.138150
Private, both correct	0.970736	0.144659	0.970736	0.144659

The parameters are those of Figure 1. A planner label records the two correction decisions. The planner optimizes both institutions' archives.

Table 2 separates the deployment and expenditure effects across the parameter combinations. An archive-information loss above 10^{-8} occurs in 21 of 81 combinations; the remaining 60 have no gap above that precision. The maximum gap is approximately 0.001380568. Figure 2 shows how the loss varies with the relative learning rates. With $\kappa = 0$, private deployment minimizes the same normalized error as the planner, as derived in Section 3.

5.3 When correction uses excessive resources

Costly correction can be privately attractive beyond its contribution to real investment. Table 3 reports the thresholds for the illustration. The private profile with both institutions purchasing correction is supported up to approximately $C_{11} = 0.007254174$. Its net value falls below that of the feasible uncorrected profile once $c > C_W = 0.002768226$. The conservative threshold for a zero-correction global planner is lower, approximately $C_P = 0.002758598$. Consequently, every interior cost between C_W and the C_{11} bound satisfies the excess-correction condition.

Table 2. Archive losses and excessive correction across environments.

b	κ	Cases	Archive gap	Excess correction
0.0	0.0	9	0	0
0.0	0.2	9	0	0
0.0	0.8	9	0	0
0.5	0.0	9	0	9
0.5	0.2	9	5	9
0.5	0.8	9	7	9
1.0	0.0	9	0	9
1.0	0.2	9	4	9
1.0	0.8	9	5	9

Each row contains all nine learning-rate pairs. "Archive gap" counts an information gap with lower bound above 10^{-8} for $J_0^* - J_{00}$. "Excess correction" counts a conservative interval of width greater than 10^{-10} on which a private profile with both institutions purchasing correction is supported and the global planner uses zero corrections.



Figure 2. Price-information loss from private archive choice. Entries are the nonnegative lower bounds for $10^4(J_0^* - J_{00})$, rounded to two decimals. Each panel contains all useful and nuisance learning-rate combinations for its displayed b and κ. The planner chooses both archives freely. The omitted $b = 0$ panels contain 27 zero-nuisance parameter combinations; the computational supplement reports each parameter combination.

Table 3. Information values and correction-cost thresholds.

Object	Value	Meaning
J_{00}	0.605577266	Private uncorrected price information
J_{11}	0.616650170	Private corrected price information
J_0^*	0.605983499	Efficient allocation with zero corrections
J_1^*	0.611500695	Efficient allocation with one correction
J_2^*	0.616660728	Efficient allocation with two corrections
C_{00}	0.007449622	No-correction equilibrium cost boundary
C_{11}	0.007254174	Both-correction equilibrium cost boundary
C_W	0.002768226	Net-value break-even cost
C_P	0.002758598	Zero-correction planner threshold

Values refer to the unequal-speed illustration with $U = \gamma = 1$. Planner values are feasible lower bounds. The displayed C_{00} uses a sufficient upper endpoint, C_{11} a sufficient lower endpoint, and C_P a sufficient upper endpoint. Values are rounded for presentation; the exact interval endpoints in the supplement govern the inequalities.

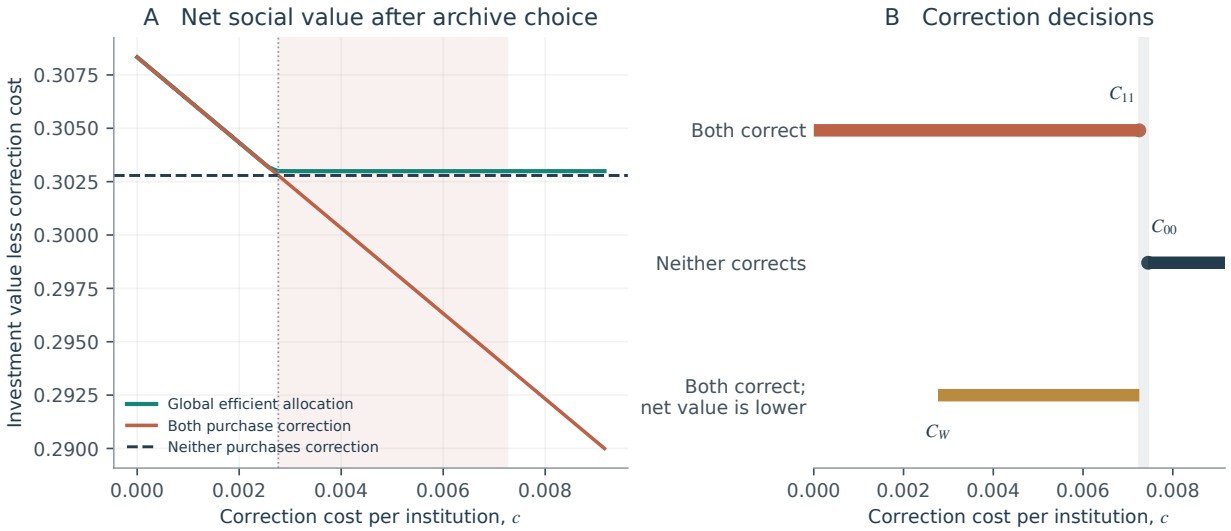


Figure 3. Private correction and the planner's net-value envelope. Panel A compares the private corrected and uncorrected profiles with the global envelope over zero, one and two corrections. Panel B marks sufficient cost intervals for the two symmetric pure regimes. The interval between C_{11} and C_{00} is outside the two symmetric pure regimes. In the shaded cost interval, the corrected private profile raises price information while consuming more resources than its real-value gain.

Figure 3 compares the private regimes with efficient expenditure. The planner's envelope includes a single corrected forecast and asymmetric archives. The interval $C_{11} < c < C_{00}$ lies between the two symmetric pure regimes; the figure leaves private equilibrium selection there open.

The expenditure effect persists with independent nuisance errors. All 54 combinations with $b > 0$ have a cost interval on which both institutions purchase correction and the planner purchases none; 18 of these have $\kappa = 0$. For $b = 0$, correction has no information benefit. This pattern separates the deployment role of correlation from the private-social valuation of a correction service.

At the midpoint $c = (C_W + C_{11})/2$, cost is 0.005011200 per institution. The corrected private profile then has net value 0.004485948 below the uncorrected private profile, or approximately 1.482 percent of

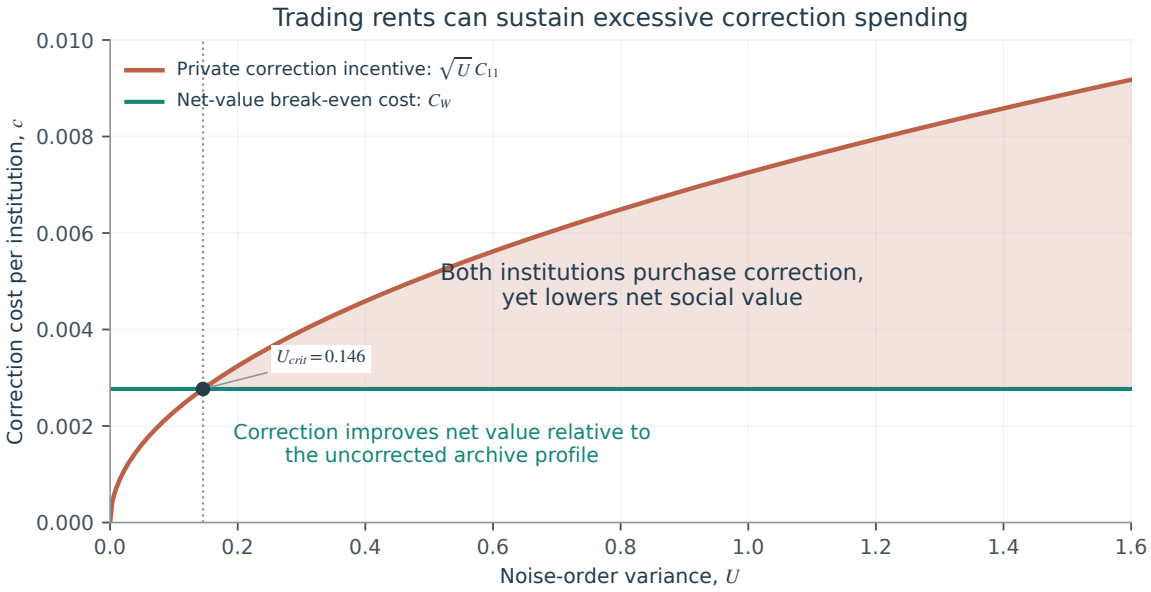


Figure 4. Noise-order variance and the private incentive to purchase correction. The economy has $b = 1$, $h_x = 1/2$, $h_w = 1/20$, $\kappa = 4/5$ and $\gamma = 1$. The upper curve is the private boundary $\sqrt{U}C_{11}(1)$. The horizontal line is C_W, which exceeds the zero-correction planner threshold C_P in this economy. Interior points of the shaded region support a profile with both institutions purchasing correction with lower net value and a zero-correction planner.

that profile's investment value. Figure 4 shows how noise-order variance changes the cost range while the real-value threshold remains fixed.

The crossing in Figure 4 is the threshold in equation (37). Above it, an interior range of correction costs supports private expenditure above the planner's choice.

6 Information access

An archive has value both through the mixture deployed and through the information accessible from its outputs. The single-output contract makes the mixture a choice of one private signal law. Access to several future realizations can instead combine the information directions of different versions.

6.1 Access to several archived outputs

Consider a joint law for the outputs of the uncorrected versions:

$$Y_{it} = x_t v + w_t^0 \zeta_i + \sqrt{d}\,\epsilon_i. \tag{39}$$

The same institution-specific residual and nuisance shock enter each version. Every nonnegative unit-sum mixture in this law has exactly the single-output distribution in equation (5). It is therefore compatible with the maintained forecast law. Its additional content is the joint distribution of forecasts that the single-output institution never observes together.

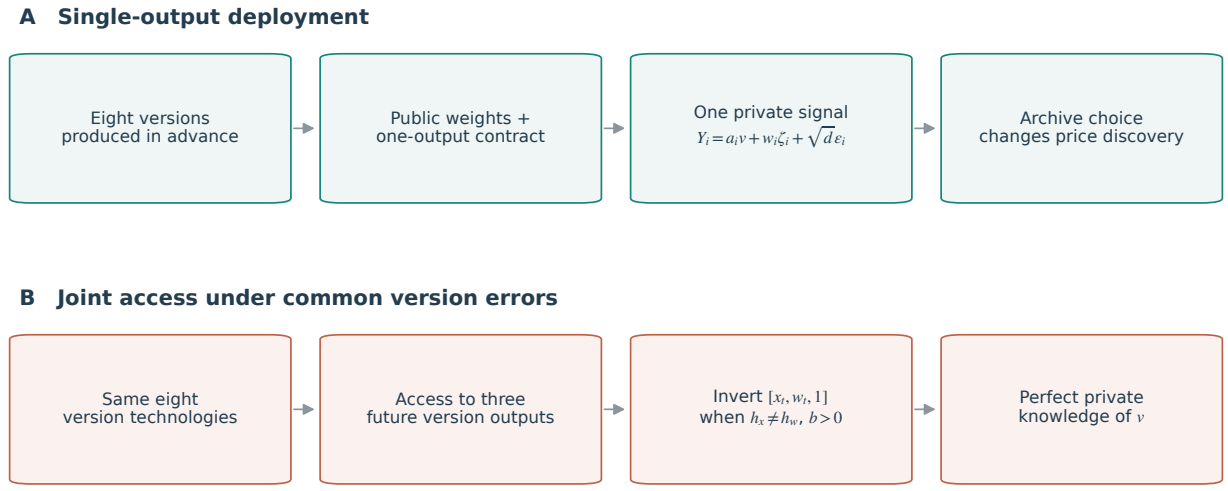


Figure 5. The deployment contract and a compatible output-access alternative. The upper branch is the maintained public single-output institution. The lower branch permits three future version realizations with common cross-version residuals, as in equation (39). Under the stated unequal-speed conditions, the loading matrix can be inverted to recover the fundamental. The figure compares information institutions using the same version technologies.

Proposition 5 (Information from multiple archived outputs)**.** *Under equation* (39)*, suppose* $T \geq 3$*,* $b > 0$ *and the positive learning rates satisfy* $h_x \neq h_w$*. Observing the first three version outputs reveals the fundamental* v *exactly.*

The archive determinant in equation (24) makes the system invertible in the fundamental and the two errors. The institution can eliminate those errors by combining the three observed realizations. This inversion uses the common cross-version errors in equation (39) and can require negative coefficients.

Perfect private knowledge still leaves noisy strategic price discovery. If all N institutions observe v, Proposition 1 gives

$$J_{\text{full}} = \frac{NV}{N+1}, \qquad \Pi_i^{\text{full}} = \frac{\sqrt{UV}}{\sqrt{N}(N+1)}. \tag{40}$$

For the two-institution normalization, price information is $2/3$. The dealer sees aggregate orders, so perfect private knowledge does not imply a fully revealing price. Competition among commonly informed traders also governs information revelation in the dynamic setting of Holden & Subrahmanyam (1992). Here the static noisy-order market preserves a distinction between privately recovered information and the price statistic.

The unequal learning rates create distinct information directions across versions. A single convex mixture trades off those directions within one output; simultaneous access permits their joint use. Access to archived outputs determines which private information institutions can bring to the same trading mechanism.

6.2 Economic interpretation

The model studies a public contract for one deployed forecast, with nonnegative weights and a correction service of known quality. Public commitment lets the dealer price each institution's signal law. Unobservable information acquisition has a different strategic role, as Banerjee & Breon-Drish (2020) shows. The common-error law in the multiple-output comparison specifies the information available under that alternative contract.

Correction improves a private forecast, while the real investor benefits through prices. Public disclosure can instead save information-production costs and improve risk sharing (Diamond, 1985), and can change private information production (Goldstein & Yang, 2017). Its real-efficiency effects also depend on the dimension disclosed (Goldstein & Yang, 2019). These distinctions place the correction expenditure result within the wider economics of information provision.

The real-value account isolates investment that responds to price information about an exogenous fundamental. In models where investment feeds back into the traded asset, information can support manipulation (Goldstein & Guembel, 2008) and asymmetric incentives to trade on favorable or adverse news (Edmans et al., 2015). Here the quadratic investment problem supplies a direct measure of the value delivered by each archive.

Trading rents are evaluated at a given noise-order variance. Informed and liquidity traders can jointly determine trading concentration (Admati & Pfleiderer, 1988), and endogenous liquidity demand changes efficiency comparisons (Mendelson & Tunca, 2004). Within the stated market, the exact $\sqrt{U}$ scaling makes the rent component transparent: increasing the private return expands correction expenditure without changing the information used by the real investor.

7 Conclusion

Private information reaches prices through institutions' trading choices. A research archive makes the composition of that information endogenous. Useful and nuisance learning generate an attainable set of forecasts, and trading profits determine the chosen mixture. With correlated errors, the private choice can differ from the one that maximizes price informativeness. Archive geometry identifies the relevant combinations of research versions.

Correction changes both the forecast and the deployment institutions prefer. Its private value reflects trading rents, while its social value reflects the investment improvement net of resources. Institutions can therefore purchase correction at costs above its real contribution even after archive reoptimization. The independent-error benchmark distinguishes this expenditure effect from inefficient archive composition.

Information technology affects price discovery through the signals institutions choose to trade on. The value of an archive therefore depends on its deployment contract and on how private trading returns compare with the investment value of prices.

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A Proofs and algebraic details

A.1 Research recursion and archive dimension

The update in equation (2) follows from a concave optimization problem. Conditional on the sample projections, define

$$\mathscr{L}_t(x, w) = x + bw - \frac{x^2 + w^2}{2} - \frac{(x - x_{t-1})^2}{2h_x} - \frac{(w - w^0_{t-1})^2}{2h_w}. \tag{41}$$

The Hessian is diagonal with entries $-1-1/h_x$ and $-1-1/h_w$, which are strictly negative. Solving the first-order conditions gives equation (2). Iterating the deviations from the fixed points gives $1 - x_t = r_x(1 - x_{t-1})$ and $b - w^0_t = r_w(b - w^0_{t-1})$. The initial conditions yield equation (3). This use of an objective plus a quadratic distance penalty is the proximal-point construction studied by Rockafellar (1976). Strict concavity establishes the update without an optimization approximation.

Equal rates collapse the archive exactly. Substituting $r_x = r_w$ into equation (3) gives the common coordinate z_t. A convex combination preserves the affine relation $(a, w) = (1/2 + z/2, bz)$, so the attainable set is the entire interval between its extreme coordinates. An endpoint mixture implements any such z. This is an equality of signal laws; it is not an approximation that assigns an average age to a two-dimensional archive.

For unequal rates, subtract the first row from the second and third rows of the determinant in equation (24). The two useful-coordinate differences factor as $\frac{1}{2}r_x(1-r_x)$ and $\frac{1}{2}r_x(1-r_x)(1+r_x)$. The nuisance-coordinate differences factor as $br_w(1 - r_w)$ and $br_w(1 - r_w)(1 + r_w)$. The remaining determinant is proportional to $r_w - r_x$, which gives the displayed formula. With positive unequal rates and $b > 0$, all its factors are nonzero.

Three versions suffice to represent any two-dimensional hull point. Suppose at least four points have positive weights. Their augmented vectors $(1, x_t, w_t)$ in $\mathbb{R}^3$ are linearly dependent. Hence there is a nonzero perturbation of those weights that preserves both coefficients and their sum. The perturbation has positive and negative entries because its entries sum to zero. Move along it until one positive weight reaches zero, retaining nonnegativity. Repetition leaves at most three points. On a supporting boundary line, every positively weighted point must lie on the exposed face, whose two endpoints span the face. Thus boundary implementation needs at most two versions.

A.2 Proof of Proposition 1

The trader's first-order condition includes the inference about rival orders. Conditional Gaussian means satisfy $\mathbb{E}[v \mid Y_i] = g_i Y_i/\Sigma_{ii}$ and $\mathbb{E}[Y_j \mid Y_i] = \Sigma_{ij} Y_i/\Sigma_{ii}$. At price rule $p = \lambda Q$, institution i chooses an order x to maximize

$$x\left(\frac{g_i}{\Sigma_{ii}} Y_i - \lambda x - \lambda \sum_{j \neq i} \beta_j \frac{\Sigma_{ij}}{\Sigma_{ii}} Y_i\right). \tag{42}$$

For $\lambda > 0$, the objective is strictly concave in x. Its maximizer is linear in Y_i, and the coefficient satisfies

$$2\lambda \Sigma_{ii} \beta_i + \lambda \sum_{j \neq i} \Sigma_{ij} \beta_j = g_i. \tag{43}$$

Collecting these equations gives $H(\lambda\beta) = g$. Since H is positive definite, $k = \lambda\beta = H^{-1}g$ is uniquely determined. The useful loadings are positive, so g is nonzero and $k'D_\Sigma k > 0$.

Dealer break-even pricing determines the remaining scale. Gaussian projection implies

$$\lambda = \frac{g'\beta}{\beta'\Sigma\beta + U} = \frac{g'k/\lambda}{k'\Sigma k/\lambda^2 + U}. \tag{44}$$

The identity $g'k = k'Hk = k'\Sigma k + k'D_\Sigma k$ then gives $\lambda^2 U = k'D_\Sigma k$. The positive square root establishes equation (7). Strict concavity verifies individual optimality, while equation (44) verifies the dealer's conditional expectation. Uniqueness here concerns the linear equilibrium with positive price impact.

Expected profit simplifies by using the first-order condition again. Unconditional profit is $\beta_i g_i - \lambda\beta_i(\Sigma\beta)_i$. Equation (43) makes this expression $\lambda\Sigma_{ii}\beta_i^2$, which is the first formula in equation (8). Moreover,

$$\mathrm{Var}(p) = \lambda^2(\beta'\Sigma\beta + U) = k'\Sigma k + k'D_\Sigma k = g'k. \tag{45}$$

Since $p = \mathbb{E}[v \mid Q]$ and $\lambda > 0$, conditioning on p is equivalent to conditioning on Q. The same expression is therefore J. Finally, maximizing the real investor's conditional quadratic payoff gives $I(p) = p/\gamma$. Its expected payoff is $\mathbb{E}[p^2]/(2\gamma) = J/(2\gamma)$, which establishes the real-value account.

A.3 Information cutoffs and unilateral derivatives

Normalization preserves both information and trading opportunities. Each a_i is positive and public, so dividing Y_i by a_i is an invertible transformation. The normalized fundamental covariance vector is $V\mathbf{1}$. At a symmetric profile,

$$\Sigma = (S-K)I_N + K\mathbf{1}\mathbf{1}', \qquad H = AI_N + K\mathbf{1}\mathbf{1}', \qquad H^{-1} = \frac{1}{A}\left(I_N - \frac{K}{D}\mathbf{1}\mathbf{1}'\right). \tag{46}$$

Thus $k = (V/D)\mathbf{1}$, and $J = NV^2/D$. Pooling gives $V^2\mathbf{1}'\Sigma^{-1}\mathbf{1} = NV^2/[S + (N-1)K]$, while one signal gives V^2/S. Substituting equation (10) proves equations (11)–(13). Each derivative has the opposite sign from the derivative of its denominator. Dividing that derivative by the positive quantity $-E'$ yields the thresholds in equation (14). Strict increase of θ orders the crossings and makes each crossing a maximum along the path.

The private derivative requires a unilateral covariance perturbation. Let e_i be institution i's coordinate vector. Write ds for its own variance change and dk_c for the change in each covariance involving that institution. Then

$$dH = 2ds\, e_i e_i' + dk_c(e_i\mathbf{1}' + \mathbf{1}e_i' - 2e_i e_i'), \qquad dD_\Sigma = ds\, e_i e_i'. \tag{47}$$

Let $k_0 = V/D$. Differentiating $Hk = V\mathbf{1}$ gives $dk = -H^{-1}(dH)k$, and equation (46) implies

$$\frac{dk_i}{k_0} = -\frac{2(D-K)ds + (N-1)(D-2K)dk_c}{AD}, \tag{48}$$

$$\frac{\mathbf{1}'dk}{k_0} = -\frac{2ds + 2(N-1)dk_c}{D}. \tag{49}$$

These expressions include the responses of all trading coefficients. They do not hold the rivals' equilibrium coefficients fixed after a public technology deviation.

To obtain the profit derivative, put $R = k'D_\Sigma k$. At symmetry $R = NSk_0^2$, and differentiation yields

$$\frac{dR}{R} = \frac{ds}{NS} + \frac{2\mathbf{1}'dk}{Nk_0} = \frac{ds}{NS} - \frac{4[ds + (N-1)dk_c]}{ND}. \tag{50}$$

Because $\Pi_i = \Sigma_{ii}k_i^2\sqrt{U/R}$, $d\log\Pi_i = ds/S + 2dk_i/k_0 - dR/(2R)$. Substituting equations (48)–(50) and collecting coefficients gives $-G_M ds - G_K dk_c$, with G_M and G_K as in equations (17)–(18). Along the archive path, $ds = E' + P'_e$ and $dk_c = \kappa P'_e/2$, which proves equation (16).

The sign argument uses no numerical parameter search. Algebraic factorization gives equations (19) and (20). Since $A, D, S > 0$, $K > 0$ and $N \geq 2$, both right-hand sides are positive. Therefore $G_M > 0$. For $\kappa > 0$, $\theta_{\text{private}} < 1$ follows from $G_K > 0$. Comparing its denominator with the price cutoff shows that $\theta_{\text{private}} > \theta_{\text{price}}$ is equivalent to $(N-1)G_M > G_K$.

At a symmetric interior pure equilibrium, the feasible own-path derivative is zero. Equation (16) therefore places $\theta(z_{\text{NE}})$ strictly between the price and standalone cutoffs. Since θ is strictly increasing, the arguments satisfy equation (22). This proves Proposition 2. The argument does not require θ_{private} itself to be increasing. For $N = 2$, the coefficients simplify to

$$G_M = \frac{12S^2 + 4SK + 3K^2}{4SAD}, \qquad G_K = \frac{1}{D}. \tag{51}$$

Multiplying the first-order condition by the positive factor $4SAD$ gives the rational polynomial equation used by the archive solver. In the equal-rate illustration, $E = 2/[5(1+z)^2]$ and $P_e = 2z^2/(1+z)^2$, so $\theta = 5z$. Exact root isolation brackets the private stationary point between the price and standalone maxima; the global deviation check then verifies the reported approximate profile.

A.4 Proof of Proposition 3

The normalized two-institution formula makes radial dominance transparent. Direct substitution in Proposition 1 gives $\Pi_i = sL^2/(D_0\sqrt{Q_0})$ and $J = 2(s+t-c_0)/D_0$. Logarithmic differentiation of the first expression gives

$$\frac{1}{s} - \frac{4t}{D_0} - \frac{L^2 + 4tM}{2Q_0} = -\frac{2c_0^2Q_0 + sD_0(L^2 + 4tM)}{2sD_0Q_0}. \tag{52}$$

The numerator on the right is positive when $L, M > 0$. Differentiating the rational expression for J gives equation (27). All denominators are positive because the signal covariance is positive definite.

An interior archive permits a proportional expansion of both coordinates. At (ra_i, rw_i), with $r > 1$ sufficiently close to one, w_i/a_i is unchanged and the rival's normalized covariance is unchanged. The own normalized variance falls because $d/(r^2a_i^2) < d/a_i^2$. Completing the square gives

$$\rho_i^2 - \frac{\kappa}{2}\rho_i\rho_j = \left(\rho_i - \frac{\kappa\rho_j}{4}\right)^2 - \frac{\kappa^2\rho_j^2}{16}, \tag{53}$$

which establishes equation (28) for both institutions. Thus the variance reduction strictly raises private profit and price information. It rules out an interior best response and an efficient allocation with an inte-

rior archive. Multiplying private profit by $\sqrt{U}$ leaves the sign unchanged for any $U > 0$. The two-version boundary representations derived above implement every remaining candidate.

A.5 Correction and information access

The joint deviation conditions follow by partitioning deviations according to correction status. At $(r_1, 1)$, a deviation within $\mathscr{H}_1$ changes gross profit only and is ruled out by conditional equilibrium. A deviation to $r \in \mathscr{H}_0$ saves c but gives gross profit no greater than $\Pi(r_1, r_1) - C_{11}$. Hence it cannot improve net profit when $c \leq C_{11}$. The no-correction argument is identical after reversing the comparison, which gives $c \geq C_{00}$.

The global planner's inequality follows from the three affine cost lines. If $c > C_P$, its zero-correction line exceeds the positive-correction alternatives, which include the line for one correction with asymmetric archives. The private-profile value difference is

$$W_{11} - W_{00} = \frac{J_{11} - J_{00}}{2\gamma} - 2c = 2(C_W - c). \tag{54}$$

Combining the private condition, the planner condition and this value difference proves Proposition 4. The conditions are strict only where the claim requires a strict welfare ranking or unique correction count. They do not require a unique private equilibrium.

For Proposition 5, collect the three observations in equation (39) into a linear system with unknown vector $(v, \zeta_i, \sqrt{d}\epsilon_i)'$. The loading matrix has rows $(x_t, w_t^0, 1)$. Its determinant equals equation (24), because a cyclic permutation of three columns preserves the determinant. The maintained unequal-speed conditions make it invertible. The fundamental is therefore measurable from the three outputs. If all institutions observe v, use the normalized signal $Y_i = v$ in Proposition 1: $H = V(I_N + \mathbf{1}\mathbf{1}')$ and $k = \mathbf{1}/(N+1)$. This yields the price information and profit in equation (40).

B Numerical implementation

The archive is constructed from exact rational coordinates. For two institutions, Proposition 3 reduces the optimization to vertices and edges. The private problem evaluates edge endpoints and isolates all stationary points of the unilateral payoff. The planner maximizes price information over pairs of edges, with asymmetric archives feasible throughout.

Numerical private profiles are ε-Nash profiles with $\varepsilon = 10^{-10}$. The largest computed unilateral-gain bound is 5.486×10^{-19}. Planner values have a maximum upper-minus-lower gap of 9.905×10^{-9}, within 10^{-8}. At a switching threshold, the best alternative archive is evaluated against the rival's fixed deployment and the resulting trading equilibrium. These conventions apply to the numerical results in Section 5.

Interval arithmetic encloses the objective values (Moore et al., 2009). Private bounds cover all stationary roots and endpoints. Rational Bernstein subdivision bounds the planner's objective on each edge pair. The computational supplement provides the algorithms, supporting calculations and the table of all 81 parameter combinations.